scRNA QC and Filtering

2026-04-17

Introduction

Single-cell RNA-seq experiments routinely contain empty droplets, stressed or apoptotic cells, doublets (two cells captured in one droplet), and various technical artefacts that should not enter downstream clustering or differential-expression analysis. Per-cell QC filtering by library size, detected-gene count, and mitochondrial-read percentage removes the worst offenders. The trade-off is delicate: over-filtering excludes real biology (e.g. plasma cells with naturally low gene counts), while under-filtering produces clusters of damaged cells that mimic real cell types.

Prerequisites

A working understanding of scRNA-seq counts, the structure of droplet-based experiments, and the role of mitochondrial reads as a cell-health indicator.

Theory

Three canonical per-cell metrics dominate QC:

- **nCount_RNA** (total UMI count per cell): very low values indicate empty droplets or debris; very high values often indicate doublets.
- **nFeature_RNA** (number of detected genes): the lowest quantile typically corresponds to broken or empty droplets.
- **percent.mt** (percentage of reads mapping to mitochondrial genes): elevated values indicate apoptotic or stressed cells whose cytoplasmic mRNA has been degraded preferentially.

Thresholds are dataset-specific; fixed cutoffs (`percent.mt < 15` for 10x human samples) work for typical experiments, but adaptive thresholds based on the median absolute deviation (`scater::isOutlier`) generalise better across datasets.

For doublet detection, dedicated tools (`scDblFinder`, `DoubletFinder`) provide more reliable identification than naive UMI-based heuristics.

Assumptions

Mitochondrial percentage is a valid proxy for cell health (fails in some tissue types — cardiomyocytes naturally have very high mitochondrial content); doublets show abnormally high UMI counts on average; the chosen filtering thresholds preserve biological diversity.

R Implementation

```r
library(Seurat)

set.seed(2026)
counts <- matrix(rpois(300 * 200, lambda = 2), 200, 300)
rownames(counts) <- c(paste0("MT-", 1:10),
                      paste0("gene_", 11:200))
colnames(counts) <- paste0("cell_", 1:300)
```

```r
counts[1:10, 1:30] <- counts[1:10, 1:30] + 20   # 30 cells with high MT

so <- CreateSeuratObject(counts)
so[["percent.mt"]] <- PercentageFeatureSet(so, pattern = "^MT-")

VlnPlot(so, c("nFeature_RNA", "nCount_RNA", "percent.mt"),
        ncol = 3, pt.size = 0.1)

so_f <- subset(so,
               subset = nFeature_RNA > 50 &
                        nFeature_RNA < 180 &
                        percent.mt   < 15)
dim(so_f)       # cells x genes after QC
```

Output & Results

`PercentageFeatureSet()` adds the mitochondrial percentage to per-cell metadata; `VlnPlot()` shows the distribution of QC metrics across cells. `subset()` applies thresholds and returns a filtered Seurat object. Inspecting the violin plots before and after filtering confirms that the chosen thresholds remove the tails without truncating the bulk of the distribution.

Interpretation

A reporting sentence: "Per-cell QC removed 4{,}166 of 12{,}400 cells (33.6 %) that fell outside the chosen thresholds (nFeature < 500, nFeature > 6{,}000, percent.mt > 15); doublet detection with `scDblFinder` flagged an additional 312 cells (3.8 %), yielding 7{,}922 high-quality cells for clustering. Thresholds were chosen by inspecting violin plots per sample." Always state thresholds and the rationale.

Practical Tips

- Inspect QC metrics per sample or batch separately before pooling; differing distributions reflect technical variation that should inform sample-specific thresholds.
- Use median-absolute-deviation-based thresholds (`scater::isOutlier(metric, nmads = 3)`) for reproducibility across datasets; fixed cutoffs are sample-specific.
- Run doublet detection (`scDblFinder` is the modern recommendation, `DoubletFinder` is the older popular alternative) separately from count-based QC; doublets often pass naïve UMI thresholds.
- Remove cells failing on multiple metrics (low UMI *and* low gene count *and* high mitochondrial percentage) rather than aggressively filtering on any single metric; combinations are more reliable than thresholds on individual metrics.
- Document QC parameters in the methods section; results are highly sensitive to filter choice and reproducibility requires explicit thresholds.
- For tissues where mitochondrial percentage is biologically meaningful (heart, skeletal muscle), adjust thresholds accordingly or use alternative QC metrics.

R Packages Used

`Seurat::PercentageFeatureSet()`, `VlnPlot()`, `subset()` for the standard QC workflow; `scater::isOutlier()` for adaptive MAD-based thresholds; `scDblFinder` and `DoubletFinder` for doublet detection; `emptyDrops` (in `DropletUtils`) for distinguishing real cells from empty droplets at the upstream step.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat